SCm_Mizuno_LENR_Transmutation

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PAPER_1140: SCm Vacuum Manifold ---\\Mizuno LENR Transmutation Mechanism

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Abstract

Mizuno LENR experiments (Pd--D and Ni--H gas-loading) report anomalous excess heat and elemental transmutation products without the hard radiation expected from nuclear reactions. We derive both phenomena from the SCm Vacuum Manifold using the $F_{U,Bi,i}$ buoyancy field, 1.25 THz phonon resonance, and the 26D Ramanujan amplification factor $S_{26}^{(3)} = 1.4531\times10^{26}$.

Canonical Constants

Constant	Value	Description
E_{phonon}	8.28×10^{-22} J	THz phonon energy
$S_{26}^{(3)}$	1.4531×10^{26}	26D Ramanujan amplification
Φ_{res}	0.84	Resonance coupling
β_i	0.6	Buoyancy coefficient
κ	5×10^{-4} day^{-1}	SCm decay rate
ρ_{SCm}	7.09×10^{-37} J/m^3	SCm vacuum density
ρ_{UA}	7.09×10^{-36} J/m^3	UA vacuum density

Mizuno Excess Heat (SCm) Mizuno observed 10--300~W excess heat in gas-loaded Ni--D/H systems with anomalous transmutation products (Cu, Cr, Fe from Ni). The SCm prediction follows the same phonon pathway as Parkhomov but with a lower cluster density N_M: $$P_{\text{Mizuno}} = N_M\cdot\varepsilon_{\text{cluster}}\cdot e^{-\kappa t}\cdot f_b$$ where $\varepsilon_{\text{cluster}}=630\ \text{eV}=1.009\times10^{-16}\ \text{J}$ is the canonical Holmlid-calibrated cluster energy, and f_b is the system-specific buoyancy stabilisation factor. For $N_M\sim10^{20}$--10^{21}: $$P_{\text{Mizuno}} \approx 10\text{--}300\ \text{W}$$

Transmutation Mechanism

Standard electroweak theory cannot explain transmutation at low temperatures without MeV-scale particle exchange. SCm provides the mechanism:

1. The 1.25 THz phonon ($E_{\text{phonon}}=8.28\times10^{-22}$ J) drives coherent oscillation of the SCm vacuum manifold within the metal lattice.
2. $F_{U,Bi,i}$ buoyancy modifies the effective nuclear potential barrier via the $\cos(\pi t_n)$ negative-time term, allowing sub-barrier transmutation.
3. The 26D Ramanujan factor $S_{26}^{(3)}$ amplifies the transition probability without requiring high-energy intermediaries.
4. Energy is released into the phonon bath (heat) rather than into particle channels (no hard radiation), consistent with Mizuno's calorimetry.

Unified LENR Picture

Experiment	System	Observed P	SCm Prediction
Holmlid	K--Fe catalyst	630 eV KER	$\varepsilon_{\text{cluster}}=630$ eV
Parkhomov	Ni--H, 1100°C	150--280 W	$N\sim2\times10^{18}$, $f_b=1$
Pons--Fleischmann	Pd--D	1--50 W	$V=10^{-6}$ m^3, $f_b=0.001$
Mizuno	Ni--D gas	10--300 W	$N\sim10^{20}$--10^{21}, f_b scaled

All four experiments are unified under a single equation (Eq.):

$$P_{\text{LENR}} = N_{\text{eff}}\cdot\varepsilon_{\text{cluster}} \cdot e^{-\kappa t}\cdot f_b(\text{system})$$

Conclusion

SCm provides a single first-principles mechanism --- phonon resonance amplified by the 26D vacuum manifold and stabilised by $F_{U,Bi,i}$ buoyancy --- that quantitatively reproduces all four major LENR experimental results without ad hoc parameters beyond those already calibrated (κ, $[SSq]$, β_i, Φ_{res}).

The unified heat equation with canonical $\varepsilon_{\text{cluster}}=630$ eV and system-appropriate N_M and f_b spans the full experimental range from Pons--Fleischmann (1--50 W) to Mizuno (10--300 W) and Parkhomov (150--280 W).